

Hrutuja Umesh Satpute

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Professional Profile

Backend Engineer with 3 years of combined experience building **scalable distributed systems** and **high-throughput** backend applications using Python, Java, and microservices architectures. Experienced in **containerized deployments**, **optimizing backend workflows** handling 28M+ transactions, and improving system reliability to 99.9% for enterprise telecom platforms. Strong foundation in **distributed systems**, **backend scalability**, **API design**, **production debugging**, **service reliability**, and software development lifecycle practices.

Experience

Amdocs - Full Time

Aug 2022 - Jul 2024

Software Developer

Pune, Maharashtra

- Designed, developed, and maintained scalable backend applications using **Python, Java, and REST APIs** by implementing efficient system design patterns, **achieving 99.9%** system reliability.
- Delivered **high-throughput solutions** for enterprise clients by optimizing data processing pipelines and API performance, handling **28M+** transactions with reduced latency.
- Containerized and orchestrated services using **Docker, Kubernetes and Helm** by automating deployment workflows and reducing infrastructure related incidents by **19%**.
- Recognized as **Employee of the Quarter** for improving project delivery timelines by **29%** through efficient system design, reducing the manual effort by using automation and cross team collaboration.
- Debugged and resolved **production issues** through root cause analysis and permanent fixes, reducing recurring incidents by **11%**.
- Conducted backend code reviews to **improve code maintainability**, **release quality**, and adhering to **engineering standards** and best practices.

UNT- Part Time

Aug 2025 – May 2026

Teaching Assistant

Denton, TX

- Assisted and managed **350+** students with Hadoop, Hive, Spark, and data analytics concepts through technical assistance.
- Supported student learning through assignment evaluation, technical mentoring, and problem-solving assistance for coursework.

Core Competencies

Distributed Systems | Backend Development | Microservices Architecture | REST API Development | Event-Driven Architecture | Backend Scalability | Cloud-Native Applications | Containerization | Asynchronous Processing | Production Debugging

Technical Skills and Certifications

Programming Languages: Python, HTML, CSS, JavaScript, React, Java

Frameworks and Libraries: Django REST Framework, Spring Boot, NumPy, Pandas, Matplotlib, Scikit-learn, LangChain, LLM

Database: SQL, MySQL, Data Integrity, PostgreSQL, SQL Server, NoSQL concepts

Platform & DevOps: Kubernetes, Helm, Docker, CI/CD (Jenkins), Kafka, Cassandra, AWS, Linux, Bitbucket, Git/GitHub, Postman

Certifications: Google Data Analytics Certificate, Software Design (Modeling with UML), CompTIA Security

Projects

Amdocs - Open Network Inventory (ONI)

Aug 2022 – Jul 2024

- Worked on ONI, a component of the **Amdocs NEO product line**, within the **OSS domain**, managing services generated by the **service orchestrator**.
- Architected and deployed **9+** Kubernetes based microservices on **AWS** environments enabling event workflows.
- Developed scalable backend solutions using **Cassandra and Kafka** to support asynchronous communication.

AI Agent Customer Prioritization Framework

May 2026

- Reduced high missed call volumes impacting customer support operations for mobile services by building an AI powered voice agent for **11K+** customer interactions.
- Improved customer response efficiency by **43%** by developing an **LLM driven prioritization** framework using sentiment analysis, urgency scoring, and business value signals to identify and escalate high priority service requests in real time.
- Decreased manual operational workload by **51%** by integrating automated call transcription, structured summaries, and centralized follow up tracking dashboards, enabling **12x** faster resolution and improving service experience.

AWS Based Serverless Financial Portfolio Management Architecture

May 2026

- Designed and presented a scalable cloud native architecture using AWS supporting **200K+ MAU** to solve fragmented financial data integration and real time analytics challenges for advisory systems.
- Built a **serverless HLD** using AWS Lambda, API Gateway, DynamoDB, Redshift, S3 and CloudFront for real time analytics.
- Evaluated **\$68K–\$70K** infrastructure costing, scalability tradeoffs, and operational efficiency improvements, leading the solution to be recognized as a **strong secondary recommendation** during technical review.

Education

Master of Science in Information Systems and Technology

GPA: 4.0

University of North Texas, Denton, TX

May 2026

Bachelor of Technology in Electronics and Telecommunications

GPA: 3.3

MKSSS Cummins College of Engineering, Pune, India

May 2022